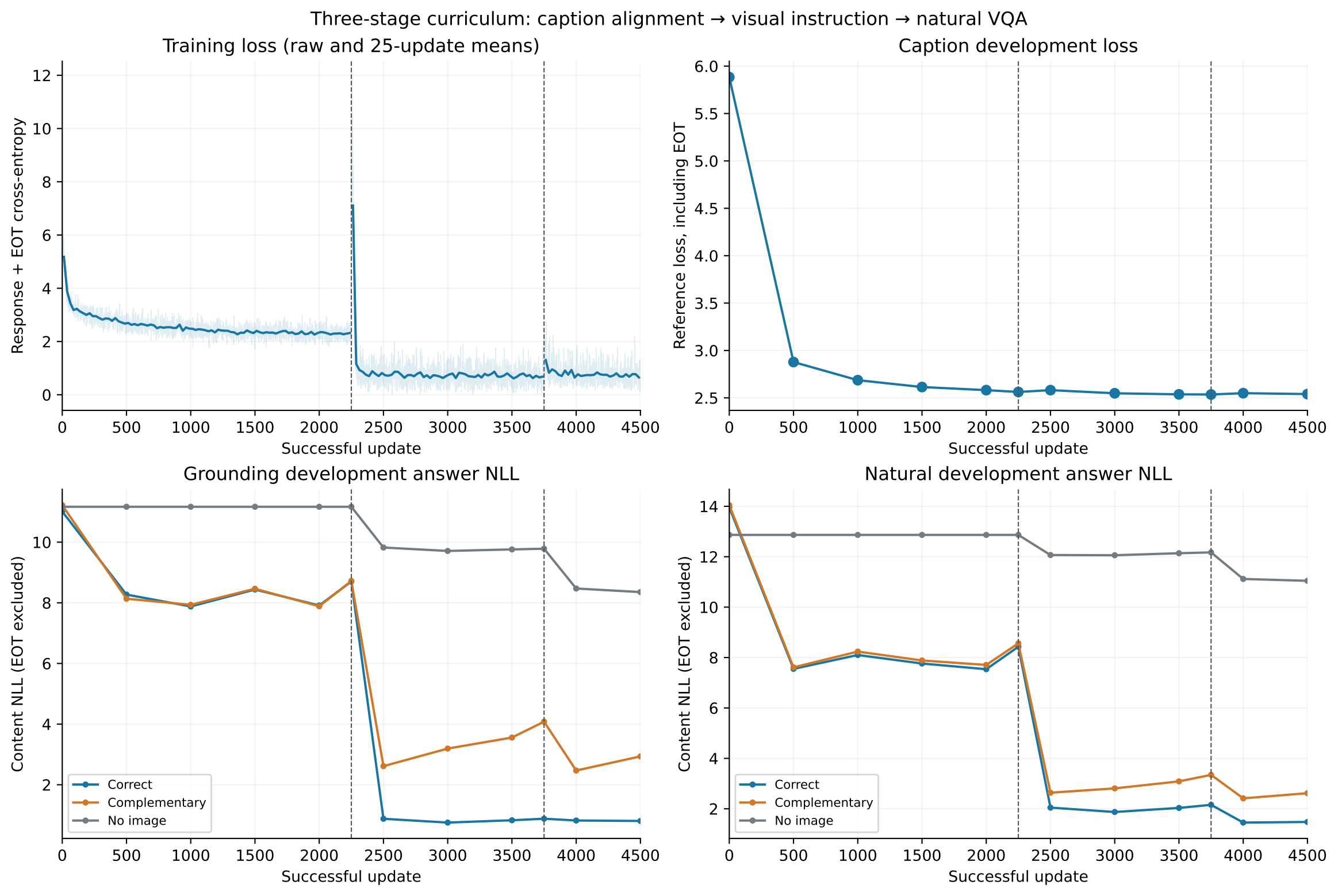

Three-stage curriculum: caption alignment → visual instruction → natural VQA
Training loss (raw and 25-update means)
Response + EOT cross-entropy
Successful update
Caption development loss
Reference loss, including EOT
Successful update
Grounding development answer NLL
Content NLL (EOT excluded)
Correct
Complementary
No image
Successful update
Natural development answer NLL
Content NLL (EOT excluded)
Correct
Complementary
No image
Successful update